

Homework 7

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Model Specification (25 points)

1. Model formulation (15 points): Write out your generalized linear mixed model. Specify the link function, the fixed effects, and the random effects structure. What assumptions are you making about the distribution of the random effects?

Our generalized linear mixed effects model is as follows:

$$g(E[Y_i|b_i]) = \beta_0 + \beta_1 x_{i,\text{YEAR}} + \beta_2 x_{i,\text{SEX}} + b_{i0} + b_{i1} x_{i,\text{YEAR}}$$

where $b_i = (b_{i0} \ b_{i1}) \sim N(0, D)$.

We use the log link function ($g(x) = \log x$). The fixed effect is $X_i \beta = \beta_0 + \beta_1 x_{i,\text{YEAR}} + \beta_2 x_{i,\text{SEX}}$. The random effect is $Z_i b_i = b_{i0} + b_{i1} x_{i,\text{YEAR}}$, which represents a random intercept and a random slope for time. We assume that $b_i \sim N(0, D)$ and $\epsilon_i \sim N(0, R_i)$ where b_i independent of ϵ_i and D and R_i cannot both be unstructured covariance matrices.

We include a random intercept to allow each subject to have their own baseline SBP level which will capture individual variation in baseline SBPs after accounting for sex and time.

Our previous analyses have also shown that systolic blood pressure increases over time at the same rate for males and for females on average, but it is plausible that on an individual level, time still has a different effect on each subject's SBP levels, so our model includes a random slope for the number of years since baseline.

2. Conditional distribution (10 points): State the conditional distribution of the response given the random effects. Explain what it means that responses are assumed independent conditional on the random effects.

Since our data is strictly positive and right-skewed, we model it using a gamma distribution.

$$\begin{aligned} f(Y_{ij}|b_i) &= \frac{\lambda^r}{\Gamma(r)} y_{ij}^{r-1} e^{-\lambda y_{ij}} \\ &= \exp((r-1) \log y_{ij} - \lambda y_{ij} + r \log \lambda - \log \Gamma(r)) \\ &= \exp\left(\frac{\frac{\lambda}{r} y_{ij} - \log \frac{\lambda}{r}}{-\frac{1}{r}} - \frac{\log r}{-\frac{1}{r}} + (r-1) \log y_{ij} - \log \Gamma(r)\right) \\ &= \exp\left(\frac{y_{ij} \theta_{ij}(b_i) - \psi(\theta_{ij}(b_i))}{a(\phi)} + c(y_{ij}, \phi)\right) \end{aligned}$$

where $\theta_{ij}(b_i) = \frac{\lambda}{r}$, $\psi(\theta_{ij}(b_i)) = \log \theta_{ij}(b_i)$, $\phi = \frac{1}{r}$, and $c(y_{ij}, \phi) = \frac{1-\phi}{\phi} \log y_{ij} - \log \Gamma(\frac{1}{\phi}) - \frac{1}{\phi} \log \phi$.

Note that, by the model's construction, all responses from the same subject i share the same random effect b_i . Said random effect captures the correlations between all responses from subject i . Therefore, given all of the random effects b_i , the responses are assumed to be independent, since all of the covariances are accounted for.

Model Fitting and Random Effects (25 points)

1. Fitting the model (10 points): Fit your GLMM using an appropriate estimation method. Report the fixed effect estimates with standard errors and the estimated variance components.

Below we fit the generalized linear mixed effects model specified in the previous section.

```
## Warning in mixed_model(fixed = SYSBP ~ SEX + YEAR, random = ~1 + YEAR | : Hessian matrix at convergence not positive definite

## Warning in sqrt(diag(var_betas)): NaNs produced

##
## Call:
## mixed_model(fixed = SYSBP ~ SEX + YEAR, random = ~1 + YEAR |
##   RANDID, data = framingham_long, family = Gamma(link = "log"))
##
## Data Descriptives:
## Number of Observations: 9618
## Number of Groups: 3206
##
## Model:
##   family: Gamma
##   link: log
##
## Fit statistics:
##      log.Lik      AIC      BIC
## -50985.81 101985.6 102028.1
##
## Random effects covariance matrix:
##              StdDev   Corr
## (Intercept)  3.1326
## YEAR         0.0774  0.9968
##
## Fixed effects:
##              Estimate Std.Err  z-value p-value
## (Intercept)   1.7231    NaN      NaN      NA
## SEX           0.0094  0.0180   0.5244 0.59999
## YEAR          -0.0712  0.0048 -14.7719 < 1e-04
##
## phi parameters:
##              Estimate Std.Err
## phi_1       5.0565  0.0287
##
## Integration:
## method: adaptive Gauss-Hermite quadrature rule
```

```
## quadrature points: 11
##
## Optimization:
## method: hybrid EM and quasi-Newton
## converged: TRUE
```

2. Random effects structure (15 points): Justify your choice of random effects structure (e.g., random intercepts only, or random intercepts and slopes). If you considered alternative random effects structures, describe how you made your selection.

Interpretation (25 points)

1. Subject-specific interpretation (15 points): Interpret your key fixed effect estimates. Be specific that these are subject-specific (conditional) interpretations. For example, for a binary outcome with logit link, interpret in terms of odds ratios for a given individual, holding the random effect constant.
2. Comparison to GEE (10 points): Compare your GLMM parameter estimates to the GEE estimates from Assignment 6. Are the estimates similar or different? Explain why the parameters from these two approaches have different interpretations.

Random Effects Predictions (15 points)

1. Predicted random effects (10 points): Obtain the predicted random effects for each subject. Create a figure displaying these predictions (e.g., a histogram or caterpillar plot).
2. Subject-specific probabilities or rates (5 points): For a few selected subjects, compute and display the predicted subject-specific response probabilities (for binary outcomes) or rates (for count outcomes) over time.

Summary (10 points)

Summarize your findings as you would for a collaborator. Address:

- What model did you fit and what random effects structure did you use?
- What are the key findings regarding your research question?
- When would you prefer GLMM over GEE (or vice versa) for your data?